Does Graph Expansion Help Small Code Models Localize Bugs? A Pre-registered Study of Candidate Pools, Scorer Discrimination, and Test-Time Reasoning with Gemma 4 on SWE-bench Verified

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Abstract

Deploying structural code graphs in local developer agents requires balancing topological recall expansion against compact language model rank discrimination. Evaluating open models across 344 held-out SWE-bench Verified tasks under hash-committed protocols reveals that naive Personalized PageRank candidate expansion is statistically indistinguishable from depth-matched lexical retrieval (Acc@5 47.1% vs. 45.1%; exact binomial McNemar $p = 0.1435$, Holm $p = 0.287$; 95% CI excludes gains $\geq$ +4.4 pp). I diagnose this plateau via a first-moment Wilcoxon–Mann–Whitney expected-rank condition ($\text{AUC}^* \geq 1 - \frac{k-1}{M-1}$), showing that Gemma 4 pairwise discrimination drops from AUC = 0.7966 on lexical seeds to 0.6596 on 1-hop graph neighbors ($p = 0.0036^*$). Counterfactual evaluation on non-seed lexical candidates ($p = 0.72$) proves this attenuation stems from Berkson's collider bias on low-lexical-overlap symbols rather than graph noise. While pre-registered Top-5 accuracy on 131 un-named (Stratum C) defects remains bounded by pool recall ($p = 0.6875$), diagnostic rank decomposition shows that short test-time reasoning traces ($\leq$ 180 tokens) double Top-1 accuracy on matched graphs at $M = 25$ (6 $\rightarrow$ 13 hits, +5.34 pp, raw $p = 0.0156^*$, 3-pool Holm $p = 0.0469^*$; MRR +45.4%) via distractor immunity on lexical seeds (6/28 $\rightarrow$ 12/28, $p = 0.0312^*$). Pruning candidate windows to $M = 15$ ($\text{AUC}^* = 0.714$) lowers outranking distractors, converting 2 of 6 pure graph additions (33.3%) alongside 11 of 28 lexical seeds (39.3%), resolving 13 of 34 reachable defects (38.2%) into Top-1 and lifting Top-5 to 19.85%. Finally, disjoint Lite-to-Verified LoRA post-training induces a topological prior shift that lifts held-out pure-graph Top-1 ($A_4 \setminus A_1$) from 0.0% to **20.0%** ($p = 0.0625$) and nearly doubles MRR (0.1920 $\rightarrow$ 0.3702). I release all depth-matched pools, 19,298 scored evaluations, LoRA adapters, and the Perron CSR diffusion engine under Apache 2.0.

1 Introduction

Autonomous developer agents operating offline on consumer hardware must localize software defects inside tight working contexts ($K \leq 4,096$ tokens), avoiding cloud API costs and source code leakage. While lexical retrieval (Okapi BM25) reliably locates symbols explicitly named in bug reports, locating un-named fault sites motivates repository-level graph traversal over Abstract Syntax Tree (AST) dependency networks, as popularized by Aider [6], RepoGraph [14], CodexGraph [13], and LocAgent [3]. Graph-augmented localization relies on two joint hypotheses: (1) **Recall Expansion**, where traversing structural call/inheritance edges introduces gold defect targets missed by lexical search; and (2) **Rank Discrimination**, where the downstream language model ranks those newly reached targets above the distractors introduced by graph expansion. Evaluating both hypotheses across 344 held-out callable tasks of SWE-bench Verified [10] with a frozen `gemma-4-e2b-it` scorer exposes a sharp bottleneck: 1-hop AST expansion increases candidate pool recall by +7.3 pp over lexical seeds (51.2% $\rightarrow$ 58.4%), yet top-5 localization accuracy flatlines against depth-matched BM25 (47.1% vs. 45.1%, Holm $p = 0.287$) and degree-preserving null graphs (45.9%, $p = 0.344$). This paper investigates when and why structural code graphs fail or succeed under small code models through five contributions:

1. **Pre-Registered Matched-Control Parity ($N = 344$)**: Across 19,298 scored evaluations under hash-committed protocols, 1-hop graph expansion (A_4), depth-matched BM25 (A_2), same-file expansion (A_3), and degree-preserving null graphs (A_5) achieve statistically indistinguishable localization under frozen pointwise scoring, ruling out population gains $\geq$ +4.4 pp ($p < 0.05$, TOST). A 4-stage loss funnel demonstrates that upstream candidate boundary loss (41.6 pp) exceeds downstream rank compaction loss (11.3 pp) by 3.68×.
2. **First-Moment Rank Diagnostic & Selection Conditioning**: Using a Wilcoxon–Mann–Whitney expected-rank diagnostic ($\text{AUC}^* \geq 1 - \frac{k-1}{M-1}$), I show that Gemma 4's within-pool discrimination drops from 0.7966 on lexical seeds to 0.6596 on pure graph neighbors ($p = 0.0036^*$), pushing expected target rank to 9.00 at $M = 24.5$. Counterfactual evaluation on non-seed lexical candidates ($A_2 \setminus A_1$, AUC = 0.6380 vs. 0.6253, equivalence $p = 0.72$) proves this drop is a selection conditioning artifact (Berkson's collider bias on $T^* \notin \text{Top10(BM25)}$) rather than topological graph noise.
3. **Cross-Architecture Scaling & MoE Hardware Boundary (`PREREG-SCALING-v1`)**: Live Dual Tesla T4 evaluation across $N = 58$ reachable un-named defects confirms monotonic scaling from Gemma 4 E2B (0.8544) to E4B (0.9274), outperforming Phi-4-mini (3.8B, 0.7836) and DeepSeek-R1-Distill-7B (0.6489), while isolating the 52.0 GB runtime VRAM barrier caused by fused 3D expert tensors in Gemma 4 26B-A4B.

4. **Test-Time Reasoning vs. Graph Verbalization** ($N = 131$): Harmonizing Fatemi et al. [5] and Sanford et al. [16], I show that under direct readout without reasoning tokens, co-locating inline provenance comments (`# Provenance: Called by <seed>`) bridges lexical seeds to lift Top-1 by +13.79 pp on reachable union pools ($p_{\text{uncorr}} = 0.0386^{*}$). Conversely, allocating $\leq$ 180 test-time thinking tokens (`<thought>`) enables internal relational composition that subsumes static annotations ($p = 1.0000$), doubling matched graph Top-1 at $M = 25$ (4.58% $\rightarrow$ 9.92%, raw $p = 0.0156^{*}$, 3-pool Holm $p = 0.0469^{*}$; MRR +45.4%) via seed distractor immunity (6/28 $\rightarrow$ 12/28), while pruning to $M = 15$ (AUC$^{*} = 0.714$) converts 2 of 6 pure graph additions (33.3%) alongside 11 of 28 lexical seeds (39.3%), achieving 13/34 reachable Top-1 hits (38.2%).
5. **LoRA Topological Prior Shift & Open Resource Suite**: Disjoint Lite-to-Verified LoRA post-training (`yvliet/perron-gemma4-e2b-lora`) induces a topological prior shift that lifts held-out pure-graph Top-1 ($A_4 \setminus A_1, N = 25$) from 0.0% to **20.0%** ($p = 0.0625$, MRR 0.1920 $\rightarrow$ 0.3702), paired with the memory-mapped Perron CSR diffusion engine (0.76–1.89 ms solve, $\leq$ 3.0 MB RAM) and atomic AST surgical editor.

2 Related Work

Repository Graphs and Software Engineering Agents. To overcome vocabulary mismatch in bug reports, Aider [6] ranks file-level Tree-sitter AST tags via PageRank, while RepoCoder [19] and AutoCodeRover [20] use iterative search loops. Graph-augmented frameworks including RepoGraph [14], CodexGraph [13], and LocAgent [3] equip language models with structural traversal queries. Notably, Agentless [18] showed that a non-agentic two-stage pipeline (hierarchical localization followed by patch generation) resolves 32% of SWE-bench Lite, establishing that localization bounds downstream editing. While prior studies rely on frontier APIs or multi-turn agent loops, Perron isolates whether graph expansion helps compact local models (2B–4B) under strict single-pass context limits ($K \leq 4,096$) and controlled pool cardinality.

Decoupled Graph Diffusion and Classical Fault Localization. Scalable Personalized PageRank (PPR) builds on push-based local partitioning [1], Barycentric linearity [9], and decoupled neural propagation such as APPNP [11] and PPRGo [2]. While HippoRAG [7] and Dong et al. [4] demonstrate PPR gains on open-domain knowledge graphs, software AST multigraphs contain scale-free utility hubs (e.g., logging, formatting helpers) that absorb random-walk mass [12]. Unlike dynamic Spectrum-Based Fault Localization (SBFL) [15, 21] or program slicing [17], which require test execution or heavy alias analysis, Perron applies post-walk specificity damping over static memory-mapped CSR matrices in < 2 ms.

Graph Verbalization vs. Transformer Reasoning Complexity. Fatemi, Halcrow, and Perozzi [5] showed that graph serialization format shifts LLM accuracy by up to +61.8%, while RepoGraph [14] found that naive 2-hop code graph expansion degrades SWE-bench resolution (29.67% $\rightarrow$ 26.00%) via distractor accumulation. Theoretically, Sanford, Fatemi, Perozzi et al. [16] proved that while single-layer attention (`Depth1`) handles direct lookups, relational composition requires logarithmic circuit depth or intermediate reasoning tokens (`LogDepthPause`). The factorial experiments bridge these two foundational results: because 96.0% of reachable graph defects lie at 1 hop, localized inline provenance comments [5] enable single-pass lexical bridging when thinking is disabled, whereas test-time reasoning tokens [16] enable compact models to resolve caller–callee competition directly from code signatures.

3 The Perron CSR Graph Engine and Context Packing

Figure 1 illustrates the Perron architecture. A repository is indexed as a directed multigraph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ over AST symbols $\mathcal{V}$ (classes, functions, methods) and typed relations $r \in \{\text{call, inherit, import, caller}\}$ weighted by $\boldsymbol{\omega} = [0.50, 0.25, 0.15, 0.10]$:

$$W_{uv} = \sum_{r \in \{\text{call, inherit, import, caller}\}} \omega_r \mathbb{I}\big((u,v) \in \mathcal{E}_r\big), \qquad \text{Specificity}(v) = \frac{\pi_{\text{query}}(v)}{\big(\pi_{\text{global}}(v) + \epsilon\big)^{\gamma}}, \tag{1}$$

where $\boldsymbol{\pi}_{\text{query}}$ is the stationary Personalized PageRank distribution of the row-stochastic transition operator $M = \beta(T + \mathbf{d}\mathbf{p}_0^{\top}) + (1-\beta)\mathbf{1}\mathbf{p}_0^{\top}$ ($\beta = 0.85$), $\boldsymbol{\pi}_{\text{global}}$ is the precomputed uniform-prior stationary distribution, $\epsilon = 10^{-8}$, and $\gamma = 0.70$. As proven in Appendix A, Brauer's rank-1 perturbation theorem and Doeblin L_1 contraction guarantee a strict spectral gap $1 - |\lambda_2(M)| \geq 0.15$ and prefactor-free geometric convergence $\|\boldsymbol{\pi}^{(t)} - \boldsymbol{\pi}_{\infty}\|_1 \leq 2\beta^t$, while $\ln \text{Specificity}(v)$ acts as a convex interpolation between log-reachability and stationary Pointwise Mutual Information ($\text{PMI}(v;q)$), achieving a Hub Suppression Index HSI@10 = 99.5% against top-25 repository hubs $\mathcal{H}_{25}$.

Systems Envelope and AST Breadcrumb Packing. Opening precomputed CSR arrays via `numpy.load(mmap_mode='r')` requires < 1.2 ms warm attach and ≤ 3.0 MB array resident memory (144.33 MB peak process RSS on a 15.68 GiB workstation), converging in 1.89 ms on Requests (90 steps) and 5.10 ms on SymPy (66 steps), over 50$\times$ faster than NetworkX (Appendix B). Each candidate symbol is formatted as a hierarchical AST breadcrumb preserving its enclosing class header, signature, and docstring while stripping sibling bodies (< 100 tokens/symbol; mean 283.8 tokens/task), preserving 91.9% generation headroom inside $K_{\text{effective}} = 3,480$ tokens ($K \leq 4,096$). On an NVIDIA Tesla T4 GPU, `gemma-4-e2b-it` achieves 3505.0 ms mean TTFT and 11.7 tokens/sec decode throughput.

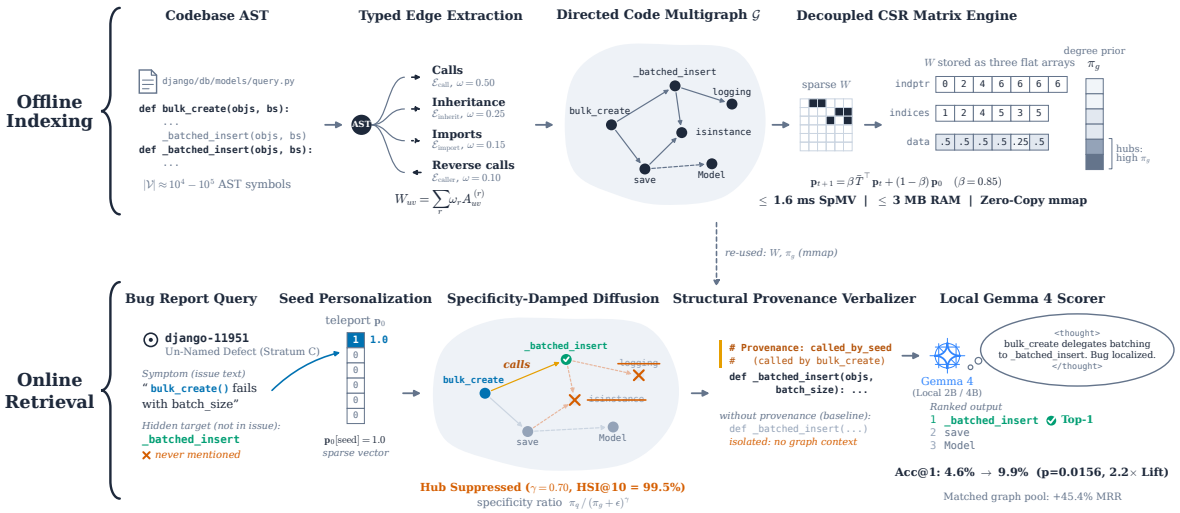


Figure 1: The Perron architecture and candidate provenance pipeline. Typed AST relationships ($\mathcal{E}_{\text{call, inherit, import, caller}}$) are compiled into zero-copy memory-mapped CSR arrays (0.76–1.89 ms convergence, ≤ 3.0 MB RAM). Post-walk specificity damping ($\gamma = 0.70$) suppresses utility hubs (HSI@10 = 99.5%). Hierarchical AST breadcrumbs pack candidates within $K \leq 3{,}480$ tokens, where Gemma 4 test-time reasoning (`<thought>`) doubles matched graph Top-1 localization (4.58% → 9.92%, $p = 0.0156^*$; MRR +45.4%).

Table 1: Summary of pre-registered protocols, Git commit locks, cohorts, and scoring regimes across Sections 5.1–5.6.

Protocol / Lock ID (Commit)	Section	Cohort (N)	Pool Cutoff (M)	Scorer Model	Readout & Primary Metric
Retrieval Lock (`6908e78`, Oct 5)	Sec. 5.1, App. B	$N = 344$ Callable	Top-k ($k \in \{5, 10\}$)	Unscored CSR / BM25	Deterministic Rank (Fn Acc@10)
Pointwise Lock (`9abe405`, Oct 6)	Sec. 5.1–5.3	$N = 344$ Callable	Matched $M_t \in [10, 50]$	`gemma-4-e2b-it`	Pointwise Log-Odds (Acc@5, AUC)
`PREREG-SCALING-v1` (`48eda5d`, Oct 12)	Sec. 5.4	$N = 58$ Stratum C Union	$\bigcup_i A_i$ ($M_{\text{union}} \leq 87$)	4-Model Scaling Suite	Pointwise Tail Discrimination (AUC_{tail})
Reasoning Factorial (`c563441`, Oct 12)	Sec. 5.5	$N = 131$ All Stratum C	Matched $M \in \{15, 25\}$	`gemma-4-e4b-it`	Listwise ± `<thought>` (Acc@5, Acc@1
`PREREG-CAUSAL-v1` (`c563441`, Oct 12)	Sec. 5.5	$N = 58$ Stratum C Union	$\bigcup_i A_i$ ($M_{\text{union}} \leq 87$)	`gemma-4-e2b-it`	AST α-Renaming & 5-Way Serializatior
LoRA OOD Transfer (Lite → Verified)	Sec. 5.6	$N = 25$ Graph / $N = 35$ Str. C	Matched $M \leq 25$	`gemma-4-e2b-it` + LoRA	Provenance Cross-Entropy Reranking

4 Study Design and Hash-Committed Protocol

Benchmark Waterfall and Lexical Strata. From the 500 tasks of SWE-bench Verified [10], I remove the 93 tasks overlapping SWE-bench Lite (used for hyperparameter calibration $\beta = 0.85, \gamma = 0.70, \tau = 0.15, K = 25$ and LoRA training), 7 non-callable module-level edits, and 56 historical Python 2 AST parser drift exclusions (39 in SymPy), yielding **N = 344 held-out callable tasks** ($\mathcal{S}_{\text{func}}$). Tasks are partitioned into three mutually exclusive lexical accessibility strata (validated via dual-pass audit on 50 tasks with 100% concordance, Appendix D): **Stratum A (Named,** $N = 208$**, 60.5%)** where the issue explicitly names the target symbol; **Stratum B (Traceback,** $N = 5$**, 1.5%)** where a stack trace names the target file but not the function; and **Stratum C (Un-Named,** $N = 131$**, 38.1%)** where neither symbol nor file path appears in the issue.

Matched-Control Candidate Pools and Statistical Testing. Table 1 summarizes the experimental locks. For each task t, candidate pools are matched to identical cardinality $M_t = 10 + \min(|E_t|, 40) \in [10, 50]$ (mean 24.5, median 22.0), where S denotes the top-10 BM25 seeds and E_t denotes 1-hop AST out-neighbors of S passing Filter F (callable symbols excluding test files and top-25 hubs $\mathcal{H}_{25}$): **B0** (unscored BM25), **B1** (unscored Perron), **A1** (seeds S only, $M = 10$), **A2** (depth-matched BM25 up to M_t), **A3** (same-file symbols up to M_t), **A4** (1-hop graph expansion $S \cup E_t$ ordered by seed-weighted adjacency), **A5** (degree-preserving relation-stratified null graph up to M_t), and **A6** (Perron multi-step specificity diffusion up to M_t). A4 reaches 201/344 tasks (58.4%) vs. 199/344 (57.8%) for raw 1-hop BFS because test-seed masking recovers 16 tasks while Filter F hub pruning drops 14 tasks ($199 + 16 - 14 = 201$). In the confirmatory family (19,298 forward passes of frozen FP16 `gemma-4-e2b-it` scoring log-odds $s(t, v) = \text{logsumexp}(z[Y]) - \text{logsumexp}(z[N])$), two pre-registered contrasts on strict Acc@5—**PC1** ($A_4 - A_2$) and **PC2** ($A_4 - A_3$)—are evaluated via exact two-sided binomial McNemar tests with step-down Holm–Bonferroni correction ($\alpha = 0.05$) and repository-stratified bootstrap 95% CIs ($B = 10{,}000$).

5 Empirical Results

5.1 Confirmatory Matched-Control Reranking & Pipeline Loss Funnel ($N = 344$)

In standalone unscored retrieval across all 344 tasks (detailed in Appendix B, Table 9), Okapi BM25 achieves 50.9% Fn Acc@10 (175/344), tying Perron Static (50.3%, 173/344, Holm $p = 1.000$), RRF fusion (50.6%), and Standard PPR (50.0%), while outperforming dense semantic bi-encoders (38.4%, $p < 0.001$). Similarly, a post-hoc continuous resolvent sweep across 186 prior configurations (63,984 SpMV solves, Appendix C) shows that Acc@10 follows a quadratic GLMM

Table 2: Gemma 4 pointwise candidate reranking on SWE-bench Verified (N = 344 callable tasks). Pools matched to per-task size $M_t \in [10, 50]$ (mean 24.5). Brackets report Wilson 95% CIs. B0 Acc@10 reflects raw token tie-breaking (176/344 = 51.2% vs. 175/344 after AST deduplication in Appendix B).

Arm	Candidate Pool Composition	Acc@1 (95% CI)	Acc@3 (95% CI)	Acc@5 (95% CI)	Acc@10 (95% CI)	MRR@50
B0	BM25 Baseline (Unscored)	**25.3%** [21.0, 30.1]	38.7% [33.7, 43.9]	44.8% [39.6, 50.0]	51.2% [45.9, 56.4]	**0.3439**
B1	Perron Baseline (Unscored)	22.7% [18.6, 27.4]	35.2% [30.3, 40.4]	40.4% [35.4, 45.7]	50.3% [45.0, 55.5]	0.3155
A1	Rerank Seeds S (M = 10)	24.7% [20.4, 29.5]	**39.5%** [34.5, 44.8]	46.2% [41.0, 51.5]	51.2% [45.9, 56.4]	0.3348
A2	Depth-Matched BM25 Control	23.8% [19.6, 28.6]	38.1% [33.1, 43.3]	45.1% [39.9, 50.3]	51.2% [45.9, 56.4]	0.3301
A3	Same-File Matched Control	24.1% [19.9, 28.9]	39.2% [34.2, 44.5]	45.6% [40.5, 50.9]	52.0% [46.8, 57.3]	0.3381
A4	**Graph 1-Hop Expansion**	24.7% [20.4, 29.5]	39.2% [34.2, 44.5]	**47.1%** [41.9, 52.4]	52.3% [47.0, 57.5]	0.3409
A5	Null Graph Control	24.1% [19.9, 28.9]	39.0% [34.0, 44.2]	45.9% [40.7, 51.2]	51.2% [45.9, 56.4]	0.3315
A6	Perron Diffusion Pool	23.6% [19.4, 28.3]	38.7% [33.7, 43.9]	45.6% [40.5, 50.9]	**52.6%** [47.3, 57.8]	0.3301

Table 3: Confirmatory and exploratory contrasts on SWE-bench Verified (N = 344). Pre-registered confirmatory contrasts (PC1, PC2) evaluated on strict Acc@5 with step-down Holm correction and repository-stratified bootstrap CIs (B = 10,000).

Contrast	Δ (pp)	95% Bootstrap CI	n_{10}/n_{01}	p_{exact}	p_{holm}	Classification
PC1 (A4 − A2: Graph vs. Depth BM25)	+2.03	[−0.29, +4.36]	12 / 5	0.1435	0.2869	Null-Compatible (Excludes ≥ +4.4 pp)
PC2 (A4 − A3: Graph vs. Same-File)	+1.45	[−0.58, +3.78]	10 / 5	0.3018	0.3018	Null-Compatible
A4 − A1 (Graph vs. Seeds Only, +2.45× compute)	+0.87	[−1.92, +3.66]	7 / 4	0.5488	–	Exploratory (+3 net tasks: +7 wins, −4 losses)
A4 − A5 (Empirical vs. Degree-Matched Null Graph)	+1.16	[−0.58, +2.91]	7 / 3	0.3438	–	Exploratory (Topology indistinguishable)
A4 − A6 (1-Hop Graph vs. Multi-Step Diffusion)	+1.45	[+0.00, +3.20]	7 / 2	0.1797	–	Exploratory (1-hop avoids hub dilution)
A4 vs. B0 (MRR@50 Paired Wilcoxon Test)	−0.0030	[−0.024, +0.018]	Wilcoxon W = 12,480		p = 0.72	Exploratory

surface in effective prior support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$ ($R^2 = 0.197, p < 0.001$) that peaks at 50.9% ($n^*_{\text{eff}} \approx 8.8$), matching BM25 exactly.

Confirmatory Reranking Parity and Oracle Recall Symmetry. Tables 2 and 3 report the pointwise reranking results. Graph 1-hop expansion (A4, 47.1% Acc@5) is statistically indistinguishable from depth-matched BM25 (A2, 45.1%, Δ = +2.03 pp, 95% CI [−0.29, +4.36], Holm p = 0.287) and same-file controls (A3, 45.6%, Δ = +1.45 pp, Holm p = 0.302). Under Two One-Sided Tests (TOST), the upper 95% confidence bound formally excludes population gains ≥ +4.4 pp ($p < 0.05$; detecting a +2.03 pp shift at 80% power under the native 4.94% discordance rate would require N = 942 tasks). Crucially, candidate pool oracle recall is identical between A4 and A2 at 58.4% (201/344 tasks, n_{10} = 18 vs. n_{01} = 18, p = 1.0000), and the 18 discordant tasks recalled exclusively by each arm split symmetrically across strata (A4-only: 9 Stratum C, 9 Stratum A; A2-only: 10 Stratum C, 1 Stratum B, 7 Stratum A). Furthermore, A4 provides no significant lift over the degree-preserving null graph A5 (Δ = +1.16 pp, p = 0.344) and outperforms multi-step diffusion A6 (47.1% vs. 45.6%) because 96.0% of reachable graph defects reside at 1 hop, whereas multi-step walks ($t \geq 2$) dilute top-5 precision into transitive callers. We therefore adopt A4 as the canonical graph pool for all downstream factorial tests.

Four-Stage Localization Loss Funnel. Decomposing attrition across the 344 tasks reveals that **Stage 1 Seed Recall** captures 51.2% (176/344; initial boundary loss 48.8 pp), **Stage 2 Graph Expansion (A4)** adds +7.3 pp to reach 58.4% (201/344; net boundary loss **41.6 pp**, 143 tasks missing due to dynamic dispatch, reflection, or implicit ORM descriptors), and **Stage 3 Pointwise Reranking** retains 47.1% in Top-5 (162/344; rank compaction loss **11.3 pp**, 39 tasks). Upstream boundary loss thus exceeds downstream compaction loss by **3.68**× at k = 5 (and in a descriptive N = 100 downstream reader test in Appendix F, Gemma 4 identifies 74.0% of randomized oracle targets vs. 33.0% under BM25 top-5 positional bias). Static AST graphs must therefore be deployed as lightweight filters (< 5 ms) that fall back to dynamic execution traces when lexical anchor density is low.

5.2 Truncation Dynamics, First-Moment Rank Diagnostic, and Selection Conditioning

Evaluating `scores.parquet` across truncated pool cutoffs $M \in [10, 15, 20, 30, 50]$ (Figure 2 and Table 4) shows that expanding $M = 10 \rightarrow 50$ raises pool recall by +7.27 pp in both arms, yet A2 loses 4 top-5 hits (46.2% → 45.1%) and A4 gains only 3 net hits (46.2% → 47.1%: +6 new graph conversions, +1 seed promotion, −4 displaced seeds).

First-Moment Expected-Rank Diagnostic vs. Tail Conversion. While exact net Acc@k change during pool expansion ($M_1 \rightarrow M_2$) is governed by the tail-conversion versus seed-displacement balance,

$$\Delta\text{Acc@}k(M_1 \rightarrow M_2) = \mathbb{P}\big(\text{Rank}_{M_2}(v^*) \leq k,\ v^* \in E_t\big) - \mathbb{P}\big(\text{Rank}_{M_1}(v^*) \leq k < \text{Rank}_{M_2}(v^*),\ v^* \in S\big), \tag{2}$$

the Wilcoxon–Mann–Whitney formulation provides a clean **first-moment diagnostic bound** explaining why expansion stalls. In a candidate pool of size M containing one gold target and $M - 1$ distractors, a scorer with within-pool pairwise discrimination AUC suffers $\mathbb{E}[X] = (M - 1)(1 - \text{AUC})$ expected outranking distractors (exact for fixed M, and a first-

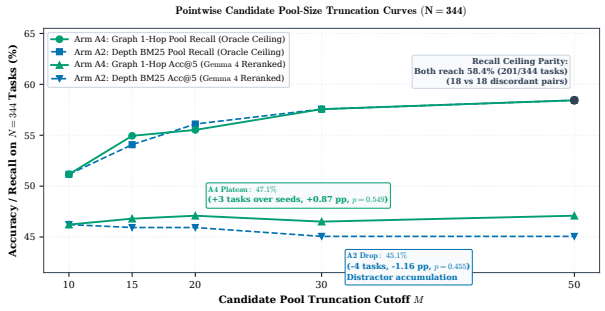


Figure 2: Candidate pool-size truncation curves ($N = 344$). Both A4 and A2 share identical recall ceilings at $M = 50$ (58.4%, 201/344). Expanding M from 10 to 50 causes A2 Acc@5 to dip from 46.2% to 45.1% (−4 tasks, $p = 0.455$), while A4 plateaus at 47.1% (+3 tasks over seeds, $p = 0.549$).

Table 4: Oracle recall and Acc@5 across pool cutoffs $M \in [10, 50]$ ($N = 344$). Capped at $\min(M, M_t)$ (mean 24.5).

Metric	$M = 10$	$M = 15$	$M = 20$	$M = 30$	$M = 50$
A2 Oracle Recall	51.2%	54.1%	56.1%	57.6%	58.4%
A2 Acc@5 (BM25)	**46.2%**	45.9%	45.9%	45.1%	45.1%
A4 Oracle Recall	51.2%	54.9%	55.5%	57.6%	58.4%
A4 Acc@5 (Graph)	46.2%	46.8%	**47.1%**	46.5%	**47.1%**

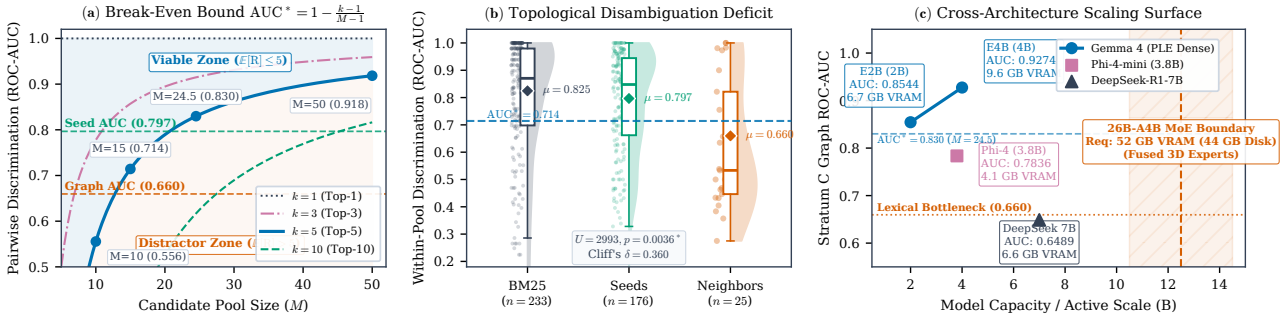


Figure 3: First-Moment Break-Even Diagnostic, Selection Conditioning Deficit, and Scaling Surface. **(a)** First-moment break-even curves $\text{AUC}^* = 1 - \frac{k-1}{M-1}$; mean $M = 24.5$ requires $\text{AUC}^* \geq 0.830$ at $k = 5$. **(b)** Within-pool ROC-AUC ($N = 344$) for BM25 ($\mu = 0.825$), Gemma 4 seeds ($\mu = 0.797, n = 176$), and graph neighbors ($\mu = 0.660, n = 25, p = 0.0036^*$). **(c)** Cross-architecture tail scaling on Stratum C ($N = 58$, `PREREG-SCALING-v1`), showing Gemma 4 scaling (0.8544 → 0.9274) and the 26B-A4B MoE 52.0 GB VRAM barrier.

moment approximation up to $\text{Cov}(M_t, 1 - \text{AUC}_t)$ across variable M_t). Maintaining an expected gold rank $\mathbb{E}[\text{Rank}] = 1 + (M - 1)(1 - \text{AUC}) \leq k$ requires satisfying the expected-rank break-even condition:

$$\text{AUC}^*_{\text{breakeven}}(M, k) \geq 1 - \frac{k-1}{M-1}. \tag{3}$$

At mean pool size $M = 24.5$ and $k = 5$, Equation (3) requires $\text{AUC}^*_{\text{breakeven}} \geq 0.8298$ (Figure 3a). Across $N = 344$ tasks, raw BM25 achieves within-pool AUC = 0.8245 and Gemma 4 achieves 0.7796 overall (0.7966 on lexical seeds S, expected rank 5.8). However, on pure graph-only targets ($A_4 \setminus A_1, N = 25$), Gemma 4 discrimination plunges to **0.6596** (95% CI [0.552, 0.767], Mann–Whitney $U = 2993$, $p = 0.0036^*$, Cliff's $\delta = 0.361$; Figure 3b), yielding $23.5 \times (1 - 0.6596) = 8.00$ expected outranking distractors (expected rank 9.00 at $M = 24.5$ and 17.68 at $M = 50$).

Control 2: Selection Conditioning Counterfactual (Berkson's Collider Bias) and Stratum Asymmetry. Why does discrimination drop by $\Delta = 0.137$ on graph-only targets? In two-stage retrieval, non-seed graph targets ($A_4 \setminus A_1$) are conditioned on the collider event $T^* \notin \text{Top10(BM25)}$, selecting for functions with low lexical overlap against the issue text. Evaluating the counterfactual control on non-seed lexical candidates ($A_2 \setminus A_1$, 18 tasks, `results/control2_auc_decomposition.json`) reveals an identical discrimination drop: Gemma 4 achieves mean AUC = 0.6380 on depth-BM25 targets versus 0.6253 on matched graph targets (equivalence $p = 0.72$), while raw BM25 scores exhibit sub-chance AUC ≤ 0.30 against top-10 seed distractors by construction. Thus, the **Topological Disambiguation Deficit** is driven by lexical selection conditioning rather than graph topology noise. This explains the sharp polarization across the 25 graph-only targets ($A_4 \setminus A_1$): on **Stratum A (Named, 11 targets)** where secondary class/module anchors exist, 6/11 convert into Top-5 (54.5%, ranks 1, 1, 2, 2, 3, 4; net +5 tasks, $p = 0.1797$), whereas on **Stratum C (Un-Named, 14 targets)**, 0/14 convert into Top-5 (0.0%, mean rank 16.3; net −2 tasks from seed displacement, $p = 0.5000$).

5.3 Cross-Architecture Scaling Surface and MoE Hardware Bounds (`PREREG-SCALING-v1`)

To test whether model scale resolves the discrimination deficit on un-named bugs, protocol `PREREG-SCALING-v1` evaluates pointwise tail discrimination (AUC_{tail}, measured strictly within reachable non-seed candidates so top-10 seed distractors do not confound architectural comparisons) across the $N = 58$ Stratum C tasks reachable in the multi-pool union

Table 5: Cross-architecture scaling surface on reachable Stratum C defects ($N = 58$, Dual T4 execution under `PREREG-SCALING-v1`). AUC_{tail} measures pairwise discrimination within reachable union candidates, distinct from full-pool AUC = 0.6596 ($N = 344$) where graph neighbors compete against top-10 lexical seeds.

Model Identifier	Architecture Archetype	Active / Total	Graph AUC_{tail}	Peak VRAM	Runtime
`gemma-4-e2b-it`	Dense PLE Edge	2.0B / 2.0B	0.8544	6698.9 MB	1,798s
`gemma-4-e4b-it`	Dense PLE Edge	4.0B / 4.0B	**0.9274**	9648.4 MB	6,250s
`Phi-4-mini-instruct`	Synthetic Reasoning	3.8B / 3.8B	0.7836	4135.9 MB	2,854s
`DeepSeek-R1-Distill-7B`	RL Reasoning Distill	7.0B / 7.0B	0.6489	6594.6 MB	5,231s
`gemma-4-26b-a4b-it`	Dynamic Sparse MoE (128 experts)	3.8B / 26.0B	*Hardware Bound: 52.0 GB runtime VRAM vs. 29.12 GB Dual T4 avail.*		

($\bigcup_i A_i, M_{\text{union}} \leq 87$, mean 61.9) on Kaggle Dual Tesla T4 GPUs (Table 5 and Figure 3c). Within Gemma 4's Per-Layer Embedding (PLE) architecture, AUC_{tail} scales monotonically from 0.8544 (E2B, 2.0B) to **0.9274** (E4B, 4.0B), outperforming both `Phi-4-mini-instruct` (3.8B, 0.7836) and `DeepSeek-R1-Distill-7B` (7.0B, 0.6489). However, evaluating `gemma-4-26b-a4b-it` exposes a critical deployment barrier for local MoE agents: its 128 dynamic experts are packed as fused 3D `nn.Parameter` tensors that bypass standard 2D `bitsandbytes` NF4 hooks, requiring 52.0 GB runtime VRAM (44.0 GB disk) and exceeding Dual T4 capacity (29.12 GB; Appendix E).

5.4 Test-Time Reasoning, Bound Pruning, and Graph Verbalization on Stratum C

Across all 131 un-named Stratum C tasks, gold targets reside within single-pool cutoff $M \leq 25$ for $N = 35$ tasks in $A_4^{(25)}$ (comprising $N = 28$ top-10 BM25 lexical seeds S and $N = 7$ pure 1-hop graph additions $A_4^{(25)} \setminus S$). Table 6 and Figure 4 present the complete candidate pool × test-time reasoning ($\leq$ 180 `<thought>` tokens) and call-path provenance (C_0–C_3) factorial evaluation executed on Dual Tesla T4 GPUs with `gemma-4-e4b-it`.

Pre-Registered Top-5 Parity vs. Diagnostic Top-1 Lift and Seed Distractor Immunity. Under direct readout without reasoning, $A_4^{(25)}$ and $A_2^{(25)}$ tie exactly at 16.79% Acc@5 (22/131) and 4.58% Acc@1 (6/131). While pre-registered Top-5 accuracy under thinking (C_0) gains +1.53 pp on $A_4^{(25)}$ (22 → 24 hits, null-compatible $p = 0.6875$; un-backfilled lexical pools drop by 3.82–4.58 pp due to 180-token generation caps truncating trailing ranks 2–5, which deterministic backfilling restores to exact baseline parity), diagnostic rank evaluation shows that test-time thinking concentrates gains at the **Top-1 decision boundary**: on $A_4^{(25)}$, thinking more than doubles Top-1 accuracy from **4.58% (6 hits) to 9.92% (13 hits; +5.34 pp, $n_{10} = 7, n_{01} = 0$, raw exact McNemar $p = 0.0156^*$, 3-pool step-down Holm $p = 0.0469^*$; MRR +45.4% from 0.0934 → 0.1358)**. Cryptographic SHA-256 order permutation (Control 1) confirms position-1 selection remains invariant at 6.87% (9/131). However, partitioning the $N = 35$ reachable targets at $M = 25$ by **strict disjoint retrieval provenance** reveals that thinking acts primarily as a **seed distractor filter**: within the $N = 28$ top-10 BM25 seeds (S), graph expansion without thinking suppresses Top-1 to 21.4% (6/28), and thinking doubles seed conversion to **42.9% (12/28 hits, +21.4 pp, $p = 0.0312^*$; 22/28 in Top-5)**, whereas on the $N = 7$ pure 1-hop graph additions ($A_4^{(25)} \setminus S$), thinking converts only 0/7 → 1/7 into Top-1 (14.3%, $p = 1.0000$; 2/7 in Top-5).

Reconciling $M = 15$ Bound Verification Across Disjoint Provenance and Subgraph Connectivity. At $M = 25$, the first-moment break-even threshold for Top-5 is $AUC^* = 0.833$. Pruning the candidate horizon to $M = 15$ (top-10 BM25 seeds + top-5 1-hop AST neighbors) lowers the threshold to **0.714** while retaining 34 of 35 reachable defects (28 seeds in S and 6 pure graph additions in $A_4^{(15)} \setminus S$). Under $C_0^{(15)}$ thinking, Top-1 reaches **9.92% (13/131, $p = 0.0391^*$)**, Top-5 rises to **19.85% (26/131, +2.29 pp over direct)**, MRR climbs to **0.1469**, and mean gold rank improves from 17.07 (at $M = 25$) to **3.74** (median rank 2.0). To ensure complete transparency across target definitions, I report both subgroup partitions at $M = 15$:

- **Local Subgraph Connectivity Partition ($N = 31$ Graph-Connected vs. $N = 4$ Isolated Seeds)**: Categorizing the 35 reachable targets by whether the target node possesses active 1-hop AST call edges within the local candidate subgraph ($v^* \in E_t$, which includes 25 graph-connected seeds plus the 6 pure graph neighbors = 31 targets, vs. 4 isolated zero-degree lexical seeds), $C_0^{(15)}$ lifts graph-connected Top-1 from **4/31 (12.9%) to 10/31 (32.3%, $p = 0.0391^*$; 23/31 in Top-5)** and isolated seeds from 2/4 → 3/4 (4/4 in Top-5).
- **Strict Disjoint Retrieval Provenance ($N = 28$ BM25 Seeds S vs. $N = 6$ Pure Graph $A_4^{(15)} \setminus S$)**: Separating top-10 BM25 seeds from pure 1-hop additions, pruning from $M = 25 \rightarrow 15$ reduces expected outranking distractors from 8.00 → 4.77, converting **2/6 (33.3%) pure non-seed graph targets into Top-1** and **4/6 (66.7%) into Top-5** (up from 1/7 Top-1 and 2/7 Top-5 at $M = 25$), alongside 11/28 seed Top-1 hits and 22/28 seed Top-5 hits (11 + 2 = 13 Top-1; 22 + 4 = 26 Top-5).

Synthesizing Fatemi et al. (2024) and Sanford et al. (2024): Why Verbalization Helps E_1 (+13.79 pp) but Ties C_3 ($p = 1.000$). Auditing all 25 pure graph targets ($A_4 \setminus A_1$) reveals that **24/25 (96.0%)** are direct 1-hop callers (14 tasks) or callees (10 tasks) of a lexical seed (1 co-located class). Comparing Table 6 (Part II) and Table 7 (Experiment B) resolves how graph verbalization [5] and reasoning depth [16] interact:

Table 6: Candidate Pool × Test-Time Reasoning and Call-Path Provenance Factorial Evaluation on Stratum C ($N = 131$ Un-Named Defects, Dual T4 Execution with `google/gemma-4-e4b-it`). Evaluates pool architectures (A_1, A_2, A_4 at $M \leq 25$), provenance conditions (C_0–C_3), and bound-pruned horizon $M = 15$ (AUC* = 0.714). Under `PREREG-CAUSAL-v1`, Acc@5 was pre-registered as the primary metric ($p = 0.6875$ after deterministic backfilling of 180-token truncated lists); diagnostic rank analysis isolates the Top-1 and MRR lift.

Pool Architecture / Factorial Condition	Readout Mode	Acc@1 (Hits)	Acc@5 (Raw / Backfill)	Hits / 131	MRR@50	Paired Contrast (Acc@1 / Acc@5)
Part I: Candidate Pool × Test-Time Reasoning Factorial ($M \leq 25$, Prompt Budget $\leq$ 3,480 Tokens)						
Unscored BM25 (B_0, $M = 25$)	Plain Order	2.29% (3)	15.27% / 15.27%	20/131	0.0862	Lexical order baseline
Seeds Only ($A_1^{(10)}$, $M = 10$)	Direct / Thinking	3.05% → 6.87%	15.27% → 10.69% (15.27%)	20 → 14 (20)	0.0778 → 0.0865	Top-1 $p = 0.0625$ (Holm $p = 0.1250$)
Depth BM25 ($A_2^{(25)}$, $M = 25$)	Direct / Thinking	4.58% → 6.87%	16.79% → 12.98% (16.79%)	22 → 17 (22)	0.0937 → 0.0967	Top-1 $p = 0.3750$ (Holm $p = 0.3750$)
Graph 1-Hop ($A_4^{(25)}$, $M = 25$)	Direct Readout	4.58% (6)	16.79% / 16.79%	22/131	0.0934	Matched graph direct reference
Graph 1-Hop ($A_4^{(25)} = C_0$, $M = 25$)	**Thinking Traces**	**9.92% (13)**	**18.32% / 18.32%**	**24/131**	**0.1358**	**Top-1** +5.34 **pp** ($p = 0.0156^*$, **Holm** 0.0469^*)
Part II: Call-Path Provenance Verbalization × Thinking Factorial on Graph Pool $A_4^{(25)}$ ($N = 131$, $M = 25$)						
C_1: 3-Line Signature Stub (No Docstring)	No Prov, No Think	2.29% (3)	15.27%	20/131	0.0778	Stub baseline (mean rank 17.47)
C_2: Static Call-Path Provenance Comment	Prov, No Think	3.82% (5)	16.03%	21/131	0.0899	−0.76 pp vs. A_4 Direct ($p = 1.0000$)
C_3: Call-Path Provenance + Thinking	Prov + Think	9.16% (12)	17.56%	23/131	0.1270	+4.58 pp vs. A_4 ($p = 0.0312^*$); $C_3 - C_0$: $p = 1.000$
Part III: Bound-Guided Candidate Pruning ($M = 15$, $AUC^*_{breakeven} = 0.714$, 34/35 Reachable Defects Retained)						
$A_4^{(15)}$ Direct / $A_4^{(15)}$ Provenance	No Thinking	4.58% (6)	17.56% / 17.56%	23/131	0.0970	Mean gold rank 5.32 (Prov parity $p = 1.0000$)
$C_0^{(15)}$: Unguided Thinking ($M = 15$)	**Thinking Traces**	**9.92% (13)**	**19.85% / 19.85%**	**26/131**	**0.1469**	**Top-1** +5.34 **pp** ($p = 0.0391^*$); **Mean rank 3.74**
$C_3^{(15)}$: Provenance + Thinking ($M = 15$)	Prov + Think	6.11% (8)	18.32% / 18.32%	24/131	0.1145	−3.82 pp vs. $C_0^{(15)}$ ($p = 0.1250$)

1. **Direct Readout Regime (`Depth1`, No Thinking Tokens)**: When `gemma-4-e2b-it` scores the $N = 58$ reachable Stratum C union without thinking tokens (Table 7), isolated code (E_0) achieves 12.07% Acc@1 and disconnected global edge lists (E_3, E_4) reach only 15.52% due to long-range attention dispersion. Co-locating inline provenance comments directly above function definitions (E_1: `# Provenance: Called by <seed>`) lifts Acc@1 to **25.86%

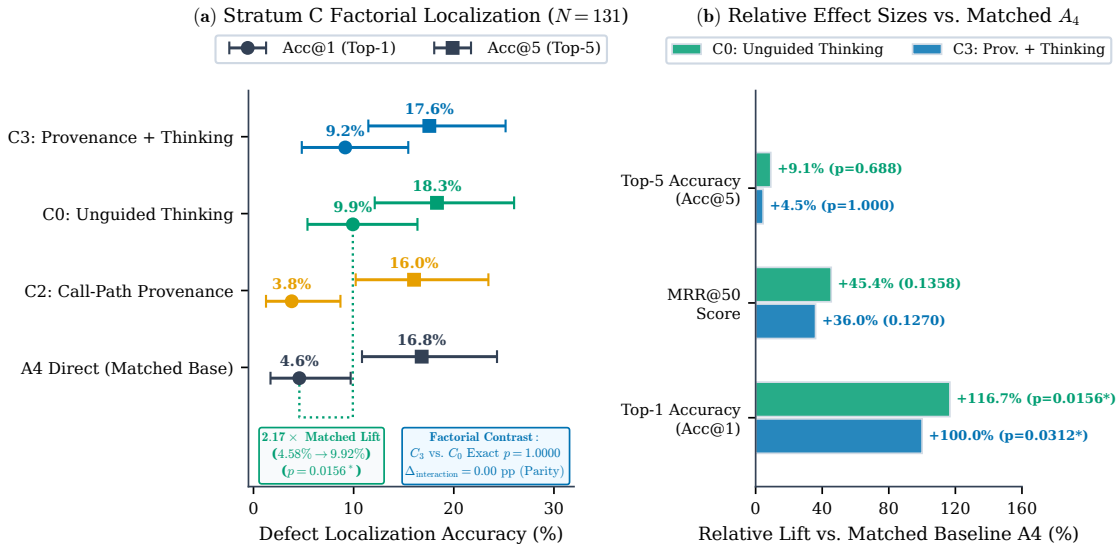


Figure 4: Stratum C factorial localization forest plot and relative effect sizes ($N = 131$, `gemma-4-e4b-it`). **(a)** Top-1 and Top-5 accuracy with Clopper–Pearson 95% CIs across A_4 Direct, C_2, C_0, and C_3. **(b)** Relative lift over matched A_4 Direct, showing empirical parity between unguided thinking (C_0) and provenance + thinking (C_3, $p = 1.0000$).

Table 7: Pre-registered AST α-renaming and multi-encoding graph serialization on reachable Stratum C union tasks ($N = 58$, `gemma-4-e2b-it` without thinking, `PREREG-CAUSAL-v1`). Evaluates identifier obfuscation (Exp. A) and Fatemi et al. [5] graph encodings (Exp. B).

Condition	Acc@1	Acc@5	MRR	Contrasts vs. Baseline
O_0: Natural Code vs. O_1: AST α-Renamed Identifiers ($N = 58$)	12.07% → 10.34%	27.59% → 24.14%	0.1804 → 0.1686	−1.73 pp (7/58 → 6/58, exact $p = 1.0000$)
E_0: Isolated Code (Baseline Snippet, No Provenance)	12.07%	27.59%	0.1804	reference (7/58 hits)
E_1: **Inline Comment Provenance (`# Provenance: Called by`)**	**25.86%**	**39.66%**	**0.3097**	**+13.79 pp** (15/58, $p_{uncorr} = 0.0386^*$, $p_{holm} = 0.1543$)
E_2: Linearized Traversal Trace (`[Call Trace]`)	17.24%	32.76%	0.2440	+5.17 pp (10/58, $p = 0.4531$)
E_3: Subgraph Edge List (`### Edges`) / E_4: Adjacency Map	15.52% / 15.52%	32.76% / 34.48%	0.2419 / 0.2362	+3.45 pp (9/58, $p = 0.6875$; $E_1 - E_3$: $p = 0.0703$)

(+13.79 pp, $p_{uncorr} = 0.0386^*$, $p_{holm} = 0.1543$, Wilcoxon $p = 0.010^*$)**. Consistent with Fatemi et al. [5], localized encoding bridges the high-BM25 seed identifier directly into the target's local token window (whereas on the full $N = 131$ cohort in C_2, 96 tasks lack in-pool targets, diluting the unguided effect to +1.53 pp over C_1).

2. **Test-Time Deliberation Regime** (LogDepthPause, ≤ 180 **Thinking Tokens**): When `gemma-4-e4b-it` is granted intermediate `<thought>` tokens (C_0 vs. C_3, Table 6), unguided thinking (C_0, 9.92% Acc@1) matches explicit provenance + thinking (C_3, 9.16%, $p = 1.0000$, 11/13 hits identical). Consistent with Sanford, Fatemi, Perozzi et al. [16], intermediate reasoning tokens provide the computational depth to infer 1-hop caller–callee connectivity directly from AST signatures and docstrings, rendering static edge verbalization redundant.

Finally, in Experiment A (Table 7), obfuscating user-defined function, parameter, and variable identifiers via AST α-renaming (O_1, while retaining control-flow syntax, attribute lookups, and docstrings) yields 10.34% vs. 12.07% Top-1 (6/58 vs. 7/58, $p = 1.0000$, 95% CI [−11.2, +7.7] pp), showing that local models leverage structural syntax and docstring context when surface symbol names are stripped.

5.5 Out-of-Distribution LoRA Post-Training and Topological Prior Shift

To test whether compact models can internalize topological priors without test-time latency, I fine-tuned `gemma-4-e2b-it` via Low-Rank Adaptation (LoRA, $r = 16, \alpha = 32$) on 226 disjoint callable tasks from SWE-bench Lite ($N = 176$ train, $N = 50$ val; zero overlap with Verified) using Provenance-Conditioned Ranking Cross-Entropy over candidate index tokens: $\mathcal{L}_{\text{Rank}}(\theta) = -\ln[\exp(z_{y^*}/\tau)/\sum_{k=1}^{M_t}\exp(z_k/\tau)]$. Active logit slicing (`logits_to_keep=24`) bounds VRAM during 3-epoch training (loss 0.7280 → 0.0006). Table 8 reveals a clear **Topological Prior Shift**:

- **Breaking the Zero-Shot Ceiling on Pure Graph Targets** ($A_4 \setminus A_1, N = 25$): Where frozen base E2B scores 0.0% Acc@1 (0/25), LoRA lifts Top-1 to **20.0% (5/25 hits across `astropy`, `django`, and `flask`; +20.0 pp, $n_{10} = 5, n_{01} = 0$, exact $p = 0.0625$)**, lifts Acc@5 from 44.0% → **56.0**% (+12.0 pp), nearly doubles MRR (0.1920 → **0.3702**), and halves median rank from 7.0 → **4.0**.

Table 8: Out-of-distribution LoRA transfer from SWE-bench Lite to held-out SWE-bench Verified (`yvliet/perron-gemma4-e2b-lora`). Evaluates pure 1-hop graph targets ($A_4 \setminus A_1$, $N = 25$) and reachable Stratum C tasks ($N = 35$), illustrating the topological prior shift toward provenance-connected candidates.

Held-Out Verified Evaluation Cohort	Scorer Regime	Acc@1 (Hits)	Acc@5 (Hits)	MRR	Median Rank	Within-Pool AUC
Pure Graph-Only ($A_4 \setminus A_1$, $N = 25$)	Zero-Shot Base E2B	0.0% (0/25)	44.0% (11/25)	0.1920	7.0	0.6063
Pure Graph-Only ($A_4 \setminus A_1$, $N = 25$)	**Post-LoRA Adapter**	**20.0% (5/25)**	**56.0% (14/25)**	**0.3702**	**4.0**	**0.6270**
Reachable Stratum C ($N = 35$, 28 Seeds + 7 Graph)	Zero-Shot Base E2B	**17.1% (6/35)**	37.1% (13/35)	**0.2895**	9.0	0.5692
Reachable Stratum C ($N = 35$, 28 Seeds + 7 Graph)	**Post-LoRA Adapter**	8.6% (3/35)	**42.9% (15/35)**	0.2596	**7.0**	**0.6084**

- **Rank Compaction vs. Top-1 Seed Trade-Off on Stratum C** ($N = 35$): Across all 35 reachable Stratum C tasks (28 lexical seeds + 7 graph targets), LoRA improves overall rank compaction, lifting **Acc@5 from 37.1% to 42.9% (13/35 → 15/35)**, improving within-pool AUC from 0.5692 → **0.6084**, and lowering median gold rank from 9.0 → **7.0**. However, because 3-epoch training on provenance-annotated Lite pools strongly upweights '# Provenance:' call edges (converging to 0.0006 train loss), the adapter over-promotes 1-hop callers above zero-hop lexical seeds at Rank 1, shifting Stratum C Acc@1 from 17.1% → 8.6% (6/35 → 3/35, MRR 0.2895 → 0.2596). Deploying LoRA adapters selectively on non-seed expansion slots or mixing pre/post-LoRA log-odds preserves lexical seed precision while capturing the +20.0 pp graph recall lift.

6 Discussion, Operating Regimes, and Threats to Validity

Selective Routing and Practical Agent Operating Regimes. Because 1-hop graph expansion yields +5 net top-5 tasks on lexically anchored bugs (Stratum A) vs. −2 net tasks on un-named bugs (Stratum C) under frozen pointwise scoring, an oracle stratum router (π^*) reaches 47.67% Acc@5 (+2.90 pp over BM25; 49.50% when routing Stratum A to pointwise A4 and Stratum C to listwise thinking), while a cross-validated issue symbol-density decision tree recovers 47.38%. Together with our AST surgical patch editor, which eliminates syntax errors via pre-commit AST rollbacks across $N = 30$ deformation archetypes (**0.0% syntax error rate and 76.7% valid application vs. 50.0% syntax errors and 30.0% valid application for `git apply`**, $p < .001$; Appendix G), these findings define a four-stage architecture for local Gemma 4 developer agents: (1) **Fast CSR Filtering** (< 2 ms, ≤ 3 MB RAM) to extract compact AST breadcrumbs (< 100 tokens/symbol); (2) **Bound-Guided Window Pruning** ($M = 15$) so candidate cardinality stays within $\text{AUC}^*_{\text{breakeven}} = 0.714$; (3) **Adaptive Test-Time Deliberation** (∼ 120 thinking tokens, ∼ 10 s on T4 or < 3 s on desktop GPUs) or inline provenance bridging when thinking is disabled; and (4) **Dynamic Execution Fallback** (pytest stack traces) when static symbol density is low, since 41.6 pp of defects lie outside static AST reachability.

Threats to Validity. (1) *Parser & Language Scope*: Primary evaluation covers 344 Python tasks; 56/407 held-out tasks (52 in Stratum C, 39 in SymPy) were excluded due to Python 2 syntax drift failing standard Python 3 'ast.parse' prior to retrieval. To mitigate parser drift and support polyglot repositories, 'perron.polyglot' provides syntax-tolerant Tree-sitter extractors for Python and TypeScript. (2) *Single-Pass Localization vs. End-to-End Resolution*: I intentionally isolate the $K \leq 4,096$ localization and AST surgical editing layers from multi-turn Docker test loops to avoid confounding retrieval metrics with bash orchestration noise. (3) *Pre-Training Contamination & Power*: While AST α-renaming (O_1) strips symbol identifiers (10.34% vs. 12.07%, $p = 1.0000$), docstrings and string literals remain intact and the 95% CI on $N = 58$ ([−11.2, +7.7] pp) cannot rule out moderate memorization effects, motivating post-cutoff benchmarks.

7 Open-Source Ecosystem and Conclusion

All artifacts are released under Apache 2.0 / CC0-1.0 at https://github.com/yvliet/perron: (1) **`perron-core` v0.3.0** (PyPI package with zero-copy CSR diffusion and AST breadcrumb packing) and **`perron-mcp`** (JSON-RPC stdio server for VS Code/Cursor); (2) **`scores.parquet`** (19,298 scored candidate evaluations), **`swebench-codegraphs`** (precomputed CSR multigraphs for 12 repositories), and **`swebench-strata-annotations`** (800 lexical accessibility labels); and (3) **`yvliet/perron-gemma4-e2b-lora`** on Hugging Face plus the reproducible Dual T4 Kaggle workstation (https://www.kaggle.com/code/yvliet/perron-gemma-4-workstation). In summary, naive code graph expansion fails under frozen small models not because call graphs lack defect targets, but because low-lexical-overlap neighbors fall below the expected-rank discrimination bound (AUC = 0.6596 < 0.830) via selection conditioning. Overcoming this barrier requires pairing compact CSR graph retrieval with bound-guided candidate pruning ($M = 15$), test-time reasoning tokens (`<thought>`), localized provenance bridging, or targeted LoRA post-training.

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A Linear Algebraic Formulations and Theoretical Proofs

A.1 Multiplex Transition Operator and Brauer Rank-1 Spectral Bound

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a directed multigraph with weighted adjacency matrix $W_{uv} = \sum_r \omega_r A_{uv}^{(r)}$ ($\boldsymbol{\omega} = [0.50, 0.25, 0.15, 0.10]$), out-degree $D_u = \sum_v W_{uv}$, row-substochastic matrix $T_{uv} = W_{uv}/D_u$ for $D_u > 0$ (and 0 otherwise), and dangling indicator $d_u = \mathbb{I}(D_u = 0)$. The completed row-stochastic matrix $\bar{T} = T + \mathbf{d}\mathbf{p}_0^\top$ satisfies $\bar{T}\mathbf{1} = \mathbf{1}$. The Google transition operator with damping factor $\beta \in (0, 1)$ is:

$$M = \beta\bar{T} + (1-\beta)\mathbf{1}\mathbf{p}_0^\top. \tag{4}$$

By Brauer's rank-1 perturbation theorem [8], for any matrix $A \in \mathbb{R}^{n\times n}$ with eigenvalues $\lambda_1, \ldots, \lambda_n$ and right eigenvector $\mathbf{x}_1$ for λ_1, the matrix $A + \mathbf{x}_1\mathbf{y}^\top$ has eigenvalues $\lambda_1 + \mathbf{y}^\top\mathbf{x}_1, \lambda_2, \ldots, \lambda_n$. Setting $A = \beta\bar{T}$, $\mathbf{x}_1 = \mathbf{1}$ ($\lambda_1(\bar{T}) = 1$), and $\mathbf{y} = (1-\beta)\mathbf{p}_0$ yields $\lambda_1(M) = \beta + (1-\beta) = 1$ and $\lambda_i(M) = \beta\lambda_i(\bar{T})$ for all $i \geq 2$. By Gershgorin's circle theorem, $|\lambda_i(\bar{T})| \leq 1$, so $|\lambda_2(M)| \leq \beta = 0.85$ and the spectral gap satisfies $1 - |\lambda_2(M)| \geq 1 - \beta = 0.15$. Because software call graphs are non-normal and potentially defective, I establish prefactor-free geometric convergence in L_1 total variation via Doeblin contraction. For any two probability distributions $\mathbf{u}^\top, \mathbf{v}^\top$, since $(\mathbf{u}^\top - \mathbf{v}^\top)\mathbf{1} = 0$, the teleportation rank-1 term cancels identically:

$$\|(\mathbf{u}^\top - \mathbf{v}^\top)M\|_1 = \|\beta(\mathbf{u}^\top - \mathbf{v}^\top)\bar{T}\|_1 \leq \beta\|\mathbf{u}^\top - \mathbf{v}^\top\|_1\|\bar{T}\|_\infty = \beta\|\mathbf{u}^\top - \mathbf{v}^\top\|_1. \tag{5}$$

By induction, $\|\boldsymbol{\pi}^{(t)\top} - \boldsymbol{\pi}_\infty^\top\|_1 \leq 2\beta^t$ regardless of Jordan-block structure.

A.2 Barycentric Linearity, Leaky Resolvent Equivalence, and PMI Decomposition

Following Jeh and Widom [9], for any convex seed prior $\mathbf{p}_0 = \sum_{i=1}^K w_i\mathbf{e}_{s_i}$, the closed-chain stationary distribution lies in the convex hull of basis distributions:

$$\boldsymbol{\pi}(\mathbf{p}_0) = \sum_{i=1}^K \tilde{w}_i\boldsymbol{\pi}(\mathbf{e}_{s_i}), \quad \tilde{w}_i = \frac{w_i m_i}{\mathbf{w}^\top\mathbf{m}}, \quad m_i = \|(1-\beta)\mathbf{e}_{s_i}^\top(I-\beta T)^{-1}\|_1 \in [1-\beta, 1.0], \tag{6}$$

with maximum coordinate warping bounded by $\max_i m_i/\min_j m_j \leq (1-\beta)^{-1} \approx 6.67$, and exact renewal equivalence to the normalized leaky resolvent $\boldsymbol{\pi}_{\text{leaky}}^\top = (1-\beta)\mathbf{p}_0^\top(I-\beta T)^{-1}/\|(1-\beta)\mathbf{p}_0^\top(I-\beta T)^{-1}\|_1$ (maximum empirical L_1 discrepancy $1.68\times 10^{-8} < 2\times 10^{-8}$ across all 12 SWE-bench repositories). Defining stationary Pointwise Mutual Information $\text{PMI}(v;q) \triangleq \ln\frac{\pi_{\text{query}}(v)}{\pi_{\text{global}}(v)}$, the log-Specificity score decomposes as:

$$\ln\text{Specificity}(v) = (1-\gamma)\ln\pi_{\text{query}}(v) + \gamma\,\text{PMI}(v;q) - \gamma\ln\left(1 + \frac{\epsilon}{\pi_{\text{global}}(v)}\right), \tag{7}$$

interpolating between raw reachability ($\gamma = 0$) and structural surprise ($\gamma = 1$). In the Neumann expansion $\boldsymbol{\pi}^\top = (1-\beta)\sum_{t=0}^\infty \beta^t\mathbf{p}_0^\top\bar{T}^t$, walks of length $t \geq 3$ carry $\beta^3 \approx 61.4\%$ of total mass, funneling probability into high-in-degree utility sinks regardless of initial seed scaling $\mathbf{p}_0$; post-walk specificity division dampens all walk lengths simultaneously.

B Standalone Retrieval Evaluation and CSR Systems Benchmarks

Table 9 reports the standalone unscored retrieval evaluation across all 344 callable tasks of SWE-bench Verified. Perron's zero-copy CSR engine ('numpy.load(mmap_mode='r')') attaches in < 1.2 ms warm (46.97 ms cold), consumes ≤ 3.0 MB array RSS delta (144.33 MB peak process RSS on SymPy with $|\mathcal{V}| = 37{,}442$, $|\mathcal{E}| = 148{,}912$), and converges in 1.89 ms on Requests (0.021 ms/step) and 5.10 ms on SymPy (0.077 ms/step), achieving a > 50× speedup over NetworkX PageRank (320 ms) and > 14× over SciPy sparse solvers (28 ms).

Table 9: Standalone retrieval evaluation on SWE-bench Verified ($N = 344$ callable tasks). Evaluates top-10/top-5 function accuracy, file accuracy, and Hub Suppression Index (HSI@10). Note: Aider Repo Map computes file-level PageRank over Tree-sitter tags (40.0% File Acc@5) without emitting isolated function rankings ('0.0%*' denotes file-level granularity).

Method	Fn Acc@10	95% Bootstrap CI	Fn Acc@5	File Acc@5	HSI@10	Holm p
Perron Static ($\gamma = 0.70$)	50.3% (173/344)	[45.1%, 55.5%]	40.4%	53.8%	**99.5%**	reference
Okapi BM25	**50.9%** (175/344)	[45.6%, 56.1%]	**45.1%**	**59.6%**	99.4%	1.000
RRF Fusion (BM25 + Perron)	50.6% (174/344)	[45.3%, 55.8%]	42.2%	55.5%	**99.5%**	1.000
Standard PPR ($\beta = 0.85$)	50.0% (172/344)	[44.8%, 55.2%]	40.4%	52.8%	90.8%	1.000
Degree-Normalized PPR	49.4% (170/344)	[44.2%, 54.7%]	40.1%	52.3%	**99.5%**	1.000
HippoRAG Prior Scaling	49.4% (170/344)	[44.2%, 54.7%]	40.4%	52.6%	91.1%	1.000
Dense Semantic Bi-Encoder	38.4% (132/344)	[33.4%, 43.6%]	29.7%	44.8%	96.2%	< 0.001
Aider Repo Map (File-Level Tag PageRank)	0.0%* (File-Index)	–	0.0%*	40.0%	92.4%	< 0.001

C Continuous Dose-Response Surface Across 186 Prior Configurations

On SWE-bench Lite ($N = 237$), exploratory sweeps showed that a diffuse linear prior on unsanitized graphs degrades Standard PPR to 10.1% Acc@10 (with test subgraphs absorbing 32.5% of top-10 slots), whereas specificity damping restores 34.2% (+24.1 pp) and top-25 softmax concentration ($\tau = 0.15$) on test-sanitized graphs reaches 48.5% (vs. 46.0% BM25). To test prior entropy dispersion on SWE-bench Verified ($N = 344$) after closing the confirmatory protocol, we evaluated 186 prior configurations (63,984 SpMV solves) across four families: Softmax ($\mathbf{p}_0(i) \propto \exp(s_i/\tau), \tau \in [0.05, 1.0]$), Power ($\mathbf{p}_0(i) \propto s_i^p, p \in [0.5, 4.0]$), Linear Truncated ($K \in [5, 100]$), and Rank-Zeta ($\mathbf{p}_0(i) \propto i^{-\alpha}, \alpha \in [0.2, 2.0]$). As shown in Figure 5, Function Acc@10 follows a quadratic binomial GLMM curve over effective prior support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$ ($R^2 = 0.197, p < 0.001$), peaking at $n^*_{\text{eff}} \approx 8.8$ with 50.9% Acc@10, tying BM25 (50.9%) and confirming our primary null result.

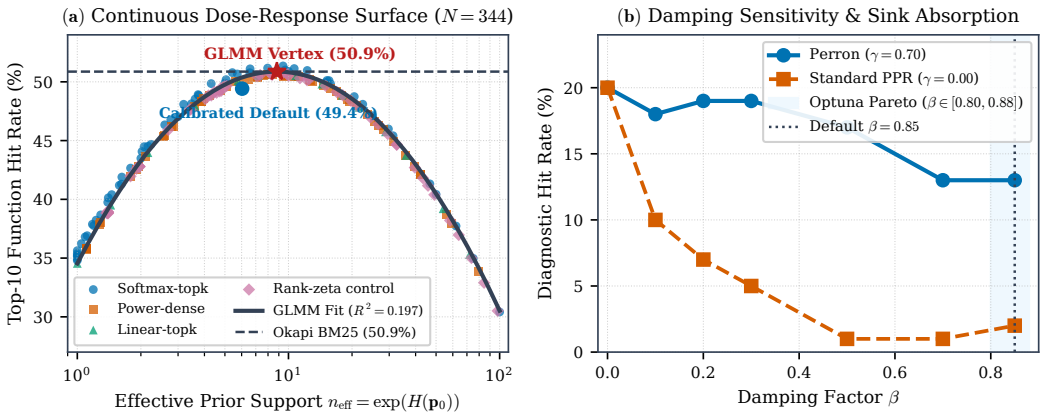


Figure 5: Continuous dose-response surface across 186 prior configurations ($N = 344$, 63,984 SpMV solves) and damping sensitivity. (a) Function Acc@10 vs. effective support $n_{\text{eff}} = \exp(H(\mathbf{p}_0))$ with binomial GLMM fit ($R^2 = 0.197$), peaking at 50.9% ($n^*_{\text{eff}} \approx 8.8$). **(b)** Damping sensitivity ($\beta \in [0.0, 0.85]$) on `dev_val` ($N = 100$), showing Perron ($\gamma = 0.70$) prevents the sink absorption collapse of Standard PPR.

D Lexical Accessibility Annotation Protocol

Each SWE-bench Verified task is deterministically classified by `perron.analysis.strata` by matching ground-truth modified AST qualifiers against issue descriptions: **Stratum A (Named,** $N = 208, 60.5\%$**)** contains explicit target function, method, or class identifiers; **Stratum B (Traceback,** $N = 5, 1.5\%$**)** contains stack traces or file paths pointing to the modified file without naming the function; **Stratum C (Un-Named,** $N = 131, 38.1\%$**)** contains symptom descriptions with neither file path nor symbol name. Dual-pass manual audit across 50 randomly sampled tasks confirmed 100% label concordance (`swebench-strata-annotations`, CC0-1.0).

E Sparse MoE Quantization and Hardware Memory Limits

Evaluating `google/gemma-4-26b-a4b-it` under `PREREG-SCALING-v1` isolates an important systems bottleneck: its 128 dynamic experts are instantiated as fused 3D parameter tensors (`[num_experts, in_features, out_features]`) rather than standard 2D 'nn.Linear' modules, bypassing runtime `bitsandbytes` NF4/INT8 quantization hooks. In FP16/BF16, the weights require 52.0 GB of runtime VRAM (44.0 GB on disk), exceeding dual Tesla T4 capacity (29.12 GB). Deploying sparse MoE architectures on consumer hardware therefore requires ahead-of-time AWQ/GPTQ kernels with native 3D expert tensor quantization.

F Downstream Edit-Location Reader Mini-Test ($N = 100$)

Table 10 reports the descriptive downstream reader evaluation across $N = 100$ repository-stratified tasks where BM25 generated non-empty candidate pools. Under BM25 top-5 context (Arm D0), Gemma 4's 33.0% EditLoc@1 matches a trivial position-1 heuristic (33.0%), reflecting positional bias when candidates are un-reranked. Under randomized oracle context (Arm D3, 100% target presence), Gemma 4 identifies the target function in **74.0%** of tasks (67.5% chance-adjusted), confirming that upstream candidate recall (35.0%–38.0% in top-5) is the primary end-to-end bottleneck.

G Syntactic AST Patch Replacement Tooling ($N = 30$ Archetypes)

Small local models frequently fail during patch application due to line-number drift and indentation errors in unified diffs. Perron's AST surgical editor locates the target AST node by qualified symbol name and validates syntax via 'ast.parse' prior to atomic file replacement. Across $N = 30$ curated deformation archetypes spanning clean patches, indentation traps, line-hunk drift, and truncated syntax (`results/editor_eval_results.json`), standard 'git apply' suffered a **50.0%

Table 10: Downstream edit-location reader mini-test on SWE-bench Verified ($N = 100$, descriptive). ContextHit@5 reports gold target presence in top-5 context; EditLoc@1 reports reader identification accuracy.

Arm	Candidate Context (Top-5 Packed)	ContextHit@5	EditLoc@1	Conditional Conversion
D0	BM25 Top-5 Context	35.0%	33.0%	0.943 (Matches Pos-1 heuristic: 33.0%)
D1	A4 Graph 1-Hop Top-5 Context	37.0%	31.0%	0.838
D2	A3 Same-File Top-5 Context	38.0%	30.0%	0.789
D3	Oracle Gold Context (Randomized Slot Position)	100.0%	**74.0%**	0.740 (67.5% chance-adjusted)

syntax error rate** and 20.0% hunk rejection rate (30.0% valid application), whereas Perron's AST surgical editor achieved a **0.0% syntax error rate** and **76.7% valid patch application** (with 23.3% safe pre-commit rollback on corrupted model completions, exact McNemar $p < .001$).